

GIANT COVALENT STRUCTURES

KEY VOCABULARY

Giant covalent: a huge lattice of atoms joined by covalent bonds.

Diamond: carbon, each atom bonded to 4 others — very hard.

Graphite: carbon in layers, each atom bonded to 3 — conducts.

Silicon dioxide: giant covalent structure, like sand/quartz.

COMMON MISCONCEPTIONS

~~—Giant covalent substances melt easily.~~

✓ They have very high melting points (strong bonds).

~~—Diamond conducts electricity.~~

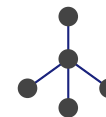
✓ Diamond has no free electrons, so it doesn't conduct.

~~—Graphite's layers are held by strong bonds.~~

✓ Layers are held by weak forces, so they slide.

DIAMOND · GRAPHITE

DIAMOND



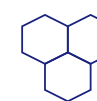
4 bonds each — hard

GRAPHITE



3 bonds — layers slide

GRAPHENE



single layer — strong

1 HIGH MELTING POINT

Explain why giant covalent structures have very high melting points. [2 marks]

2 DIAMOND HARDNESS

Explain why diamond is so hard. [2 marks]

3 GRAPHITE CONDUCTS

Explain why graphite conducts electricity but diamond does not. [3 marks]

4 SOFT GRAPHITE

Explain why graphite is soft and slippery. [2 marks]

5 FILL THE GAPS

Complete the sentence using the word bank. [2 marks]

In graphite each carbon bonds to ____ others, leaving one ____ electron to carry charge.

four

three

free

fixed

6 USES

Suggest a use for diamond and a use for graphite, linked to their properties. [2 marks]
